

DHRUV CHHABRA

✉ dhruv.c157@gmail.com ☎ +91 9810646799 📍 Delhi, INDIA 🌐 dhruwanda.github.io 📄 /dhruwanda

WORK EXPERIENCE

Quantitative Trader - Commodities; Alpha Alternatives Holdings Pvt Ltd

May 2024 – Sept 2025

- **Multi-Asset Pairs Trading & Risk Framework**

- * Generated ₹20 lakh+ P&L in 2025 on a ₹6 Cr book at a **1.9 Profit Factor** via early mean-reversion strategy
- * Protected the book from tail-risk losses via volatility-adjusted stop-losses, holding a **45% win rate** profitably
- * Partnered with discretionary traders to convert trade intuition into systematic **gradient-based momentum signals**
- * Delivered the project end-to-end after the intern's exit by auditing and rebuilding it into a systematic signal framework

- **Algorithmic Strategy Refinement and Portfolio Optimisation system**

- * Achieved **1.38% net monthly return** at a 1.2 Sharpe by rebalancing weights via Markowitz mean-variance optimisation
- * **Reduced operational errors and slippage cost ~7x** (0.15% to 0.02%) by automating signal generation
- * Prevented deployment of flawed strategies by auditing a trend-following Futures algorithm, rectifying backtesting biases

Associate Data Scientist; Optum Global Solutions Ltd

Aug 2021 – May 2024

- **Automated Clinical Documentation & Validation System**

- * Cut clinical documentation time by **55-60%** with a Gen AI engine automating medical claim-appeal summaries
- * Built a correctness-evaluation ML model to score AI summaries, **cutting manual audit effort** by 85%
- * **Scaled the engine** to cover 73% of all medical claim-appeal categories, widening the automation footprint
- * Validated **AI non-inferiority** to clinician summaries by designing a double-blind experiment for a 2-sample t-test

- **Prediction of Risk of Hospital Readmission**

- * Built a **\$10-20M projected-savings case** for cutting avoidable 30-day readmissions via targeted patient intervention
- * **Doubled the hit rate** of intervention targeting (50% precision vs 23% base rate) via a readmission risk model
- * Presented the **cost-benefit analysis** and risk drivers to the division's research director, shaping the intervention strategy

- **Prediction of First Occurrence of Multiple Diseases Simultaneously**

- * Enabled early screening across 63 chronic diseases at once by designing a multi-label neural network on patient history
- * Demonstrated feasibility of single-model multi-disease prediction as a POC to the ML/AI R&D leadership

PRODUCT & STRATEGY PROJECTS

smallcase: Strategy Performance - User Execution Gap 📊📈🔗

Aug 2026

- Scoped a Zerodha-first MVP to separate strategy returns from users' actual performance, a gap validated by app reviews
- Attributed the performance gap to price, quantity and costs by reconciling user execution with strategy trades
- Added an LLM-powered Q&A layer for granular performance analysis, grounded in deterministic financial calculations

Honasa Consumer: Turnaround and Margin Teardown 📉📊

July 2026

- Found the two-year rise in EBITDA margin from 7% to a record 11.3% came from cost lines, not from pricing
- Showed the recovery is volume-led: 30% volume growth vs 27.7% revenue in Q4 FY26, with realisation per unit falling
- Showed the FY31 margin target leans on cost efficiency and mix, raising doubts on the premiumisation narrative

The Mental Health Coverage Gap in Indian Health Insurance 📊

May 2026

- Exposed a near-0% outpatient-therapy coverage gap in Indian retail health insurance by auditing best-in-class policies
- Showed outpatient mental-health cover can be economically viable via a B2B benefits platform redesign with hybrid funding

Strategic Roadmap: Protecting Margins Amidst Global Silver Shortages 📊

Apr 2026

- Traced structural EBITDA risk for Solar PV manufacturers to a cumulative silver deficit of 700+ Moz built since 2021
- Found cost pass-through structurally fails beyond \$118/oz silver by stress-testing margins across forward price scenarios
- Recommended a zero-cost collar, miner offtake deals, and silver-reduction R&D to fix costs and secure supply

Algorithmic Trading with Backtesting Analysis

Jan 2024

- Built and back-tested systematic trading strategies (alphas) using technical indicators to generate buy/sell signals
- Back-tested the results against S&P500 as the benchmark and computed the draw-downs and risk-adjusted returns

SKILLS

Python, SQL, C++, PySpark, Git, L^AT_EX, LLM/Gen AI tooling, Statistical Testing, Advanced Excel (VBA)

EDUCATION

MSc, Statistics

Indian Institute of Technology Kanpur

8.33 /10

July 2019 - May 2021

BSc, Statistics (Hons.)

PGDAV College, University of Delhi

9.47 /10

July 2016 - May 2019

KEY COURSES

Financial Statistics

Statistical Methods for Business Analytics

Statistical Simulation & Inference

Econometrics

EXTRA CURRICULAR

- Awarded the **Junior Diploma** in Hindustani Classical Music (**Violin**) from Prayag Sangeet Samiti, 2025
- Awarded **1st Best Speaker** in **Novice Debating League** organised by DebSoc IIT Kanpur, 2020